

Determination of Currents in circuit using Mesh Analysis

Aim:-

To determine the current in circuit using mesh analysis both theoretically and practically for a given DC circuit.

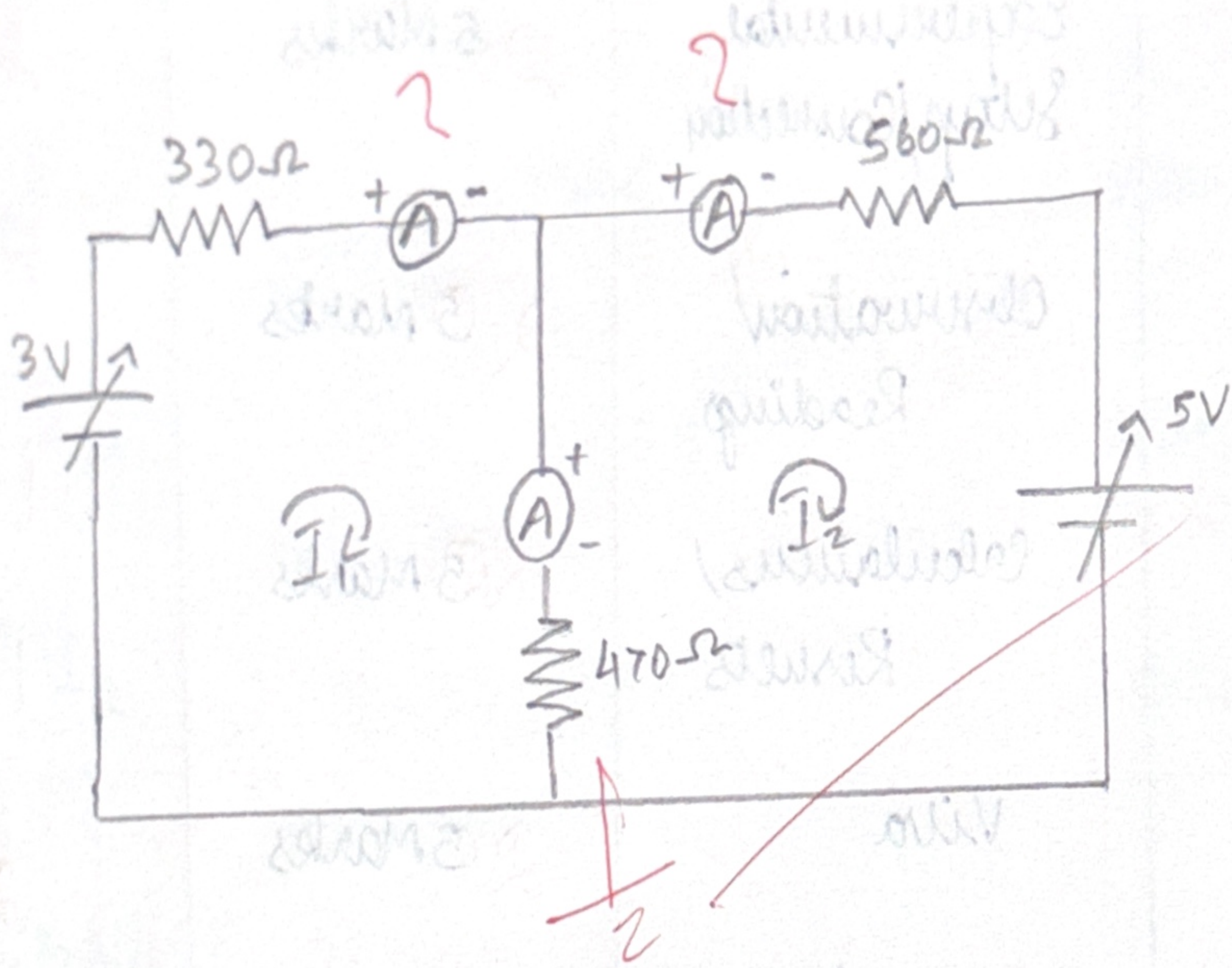
Apparatus Required:-

- | | | | |
|----|------------------------------|-------------------------------|--------|
| 1. | Regulated Power Supply (RPS) | (0-30V) | 1 |
| 2. | Ammeter | (0-10mA) MC | 3 |
| 3. | Resistors | 330- Ω , 470- Ω | Each 1 |
| 4. | Bread board | | 1 |

Procedure:-

1. Give connections as per the circuit diagram.
2. Switch ON the supply, vary the RPS.
3. Note down the readings.
4. Vary the RPS for different input voltages.
5. Reduce the RPS.
6. Using the calculated, verify Kirchhoff's law.

Circuit Diagram:-



✓

Marks Split-up:-

1.	Experimental Setup/ Connections	20M	20
2.	Observation/Readings	20M	20
3.	Calculations/Results	20M	10
4.	Viva	20M	20
5.	Record	20M	
Total:-			80

Result:-

Thus, the Mesh Analysis is verified practically and theoretically. The Resultant currents for

Supply are:-

a) 330Ω

A; 560Ω

A; 470Ω

A.

Tabular Column:-

Parameter	Theoretical	Practical
I_1	1.22	1.3mA
I_2	-4.29	-4.3mA
$I_1 - I_2$	5.51	5.3mA

Calculation:-

$$\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$R_{11} = 330 + 470 \Omega = 800 \Omega$$

$$R_{12} = 470 \Omega$$

$$R_{21} = 470 \Omega$$

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

$$\Delta = 800 \times 1030 - (470)^2 = 603100$$

$$\Delta_1 = 3090 - 2380 = 740$$

$$\Delta_2 = -4000 + 1410$$

$$I_1 - I_2 = 5.51 \text{ mA}$$